

PRODUCT SPECIFICATION DOCUMENT

Centralized Python HR Workflow Optimization Platform

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Target Budget: Absolute Zero (\$0)

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1. Executive Summary & Core Concept

The Python HR team currently operates across fragmented systems, utilizing multiple isolated Google Sheets, direct emails, and manual coordination to track candidates from registration through background verification (BGV) and eventual job placement. While functional, this spreadsheet-centric operational paradigm incurs an administrative tax of manual duplication, search inefficiencies, and data vulnerability.

This document details the specifications for a single, centralized, mobile-responsive web application designed to act as a unified "ergonomic face" over the existing data layers. Under a strict **Zero-Budget Constraint**, the application completely bypasses paid cloud infrastructure by utilizing Google Sheets natively as a secure backend database. The core business communication continues to live on **WhatsApp**, establishing an intentional boundary: WhatsApp functions as the real-time engagement mechanism, while this web application serves as the system of record.

2. Operational Scope & Platform Boundaries

To preserve developmental efficiency and avoid redundant architectures, the platform enforces hard integrations and strict functional exclusions:

Functional Area	Platform Boundary	Operational Policy & Scope Definition
Communication	WhatsApp (Irreplaceable)	All real-time team threads, immediate candidate updates, and active notifications live on WhatsApp. The web app does not attempt to replicate messaging.
Attendance & Salary	Tickora (Excluded)	No financial payroll calculations or session logging are built. Handled exclusively by Tickora.
Core Student File	Zoho (Excluded)	Detailed permanent student registration profiles remain intact inside Zoho. The web app strictly handles the operational lifecycle.
Database Layer	Google Sheets (In-Scope Backend)	All form data, profile modifications, and payment entries update the Google Sheets backend in real time, keeping storage cost-free.

3. Comprehensive Feature Specifications

3.1 The Action Trigger (Push Panel Engine)

The entry point of daily workflows is the **Push Panel**, a minimized, high-visibility input module present on the dashboard.

- **Input Parameters:** Receives a clean Candidate Email ID copied from incoming WhatsApp threads.
- **Payload Action Triggers:** A single-click tri-button interface allowing HR to trigger automated transactional workflows:
 - *Trigger A:* Dispatches an email containing a secure, unique link to the New Registration Web Form.
 - *Trigger B:* Dispatches an email containing a secure link to the detailed BGV Data Form.
 - *Trigger C:* Dispatches contextual follow-up or contact assets directly to the candidate's inbox.
- **Immediate State Change:** Upon execution, the candidate is instantly inserted into the system's Track List with an immutable visual marker labeled **Form Pending**, creating a reliable ledger of pending responses.

3.2 Automated Candidate Profiling & Ingestion

Data intake is completely offloaded to the external applicant, eradicating manual data entry routines for the core HR staff:

- Step 01** Candidate accesses the responsive web form via their mobile device or desktop browser from the triggered email link.
- Step 02** Upon form submission, the system routes the structured data parameters directly into designated cells within the background Google Sheet.
- Step 03** The web app intercepts the data synchronization, clears the **Form Pending** flag, and immediately instantiates a new interactive **Candidate Profile** on the UI.

3.3 Dynamic Search Engine & Inline Editing

The platform implements a highly responsive text-filtering framework to eliminate manual spreadsheet hunting:

- **Dynamic Filtering:** A global search interface that re-evaluates database arrays instantly on every keystroke. Candidates are filtered as HR types.
- **Multi-Dimensional Sorting:** Built-in structural toggles to filter candidate cards by Branch, Payment Delinquency, BGV Phase, or Assigned Course.
- **The Pencil Interface:** Every structured data point on the profile layout displays an inline edit pencil button. Clicking this element renders the specific field editable directly on-screen. Saving updates pushes modifications immediately to the respective Google Sheet row, protecting the document from structural formulas breaks or accidental row wipes.

Design Guardrail: Direct manipulation of rows within Google Sheets by human hands is completely abstracted away. The app functions as a defensive validation gateway for all incoming updates.

3.4 Background Verification (BGV) Pipeline

The BGV workflow transitions candidates seamlessly through structured verification gateways:

- **Initial Ingestion:** Displays a baseline status of **Form Pending** once the link is issued via the Push Panel.
- **Verification Gateway:** Once the candidate submits their comprehensive BGV data, the system advances the profile indicator to an action-oriented **Review and Send** state.
- **Vendor Hand-Off:** HR reviews the submitted verification assets directly on their dashboard screen. Clicking "Review and Send" triggers an automated backend mail server dispatch that formats, attaches, and forwards the candidate's verification package directly to the external BGV Vendor Team's inbox without requiring manual draft creation.

3.5 The Split Financial Ledger & Audit Sidebar

To prevent accounting discrepancies, the payment infrastructure utilizes isolated storage parameters with deep historical transparency:

Financial components are tracked independently via discrete mathematical logic:

$$\text{Total Fees Owed} = \text{Registration Fee} + \text{Course Tuition Fee} - \text{Total Amount Remitted}$$

The application isolates the initial *Registration Fee* separately to act as an operational gatekeeper, ensuring it is tracked distinct from ongoing tuition installments.

- **Clean Interface Display:** The primary candidate financial panel maintains a highly uncluttered presentation layer, explicitly rendering only the final aggregated tallies, active outstanding balances, and delinquent payment flags.
- **The "Changes" Audit Sidebar:** Located on the lateral edge of the payment interface is an expandable sidebar panel dedicated to tracking financial modifications. Clicking this component fetches the absolute historical record for that specific student, mapping every payment receipt, balance correction, installment entry, and staff-driven adjustment alongside a clear timestamp and editor tracking parameter.

3.6 Master Profile Lifecycle Dashboard

Clicking a candidate's identity element anywhere across the workspace opens the full layout view of their ****Master Profile****. To ensure instant operational clarity, the upper boundary of the profile view contains a row of prominent, status-driven lifecycle badges:

Milestone Badge	Selectable Parameters / States	Downstream Action & Visualization
Placement Status	In Training → Placed	Locks the operational record and updates metrics on branch-wide summary views.
BGV Phase	Form Pending → Review & Send → Cleared	Provides immediate visibility into third-party verification progression.
Corporate Entity	Text Value (e.g., [Company Name])	Maps the corporate hiring destination directly to the top-level candidate data file.

4. Technical Infrastructure (Zero-Budget Realization)

Strict Budget Constraint: No paid cloud servers, compute nodes, or database licenses are permitted. The entire application must fit within standard developer free-tiers or localized executions.

- **UI/UX Frontend:** Developed using a highly optimized, responsive JavaScript framework (e.g., React, Next.js, or HTML5/Vanilla JS) completely styled via standard layouts to guarantee fast, fluid scaling across mobile phone displays and laptop browsers.
- **Data Storage Layer:** The platform reads and writes directly from Google Workspace Sheets utilizing the free Google Sheets API or a localized middleware runner, achieving permanent cloud storage with zero infrastructure overhead.
- **Transactional Emailing:** Dispatches all forms, contact lists, and automated BGV files using a secure SMTP relay configured natively through a dedicated corporate Google Workspace/Gmail channel or a free-tier developer mail API.
- **Access Isolation:** Authentication pathways partition visibility dynamically. Branch Managers logging into the web application are structurally confined to candidate data records tagged with their respective branch location, while global administrators retain full cross-branch master visualization.